

1 Congruent Triangles

Triangles are congruent if they have the same size and shape. This means all corresponding sides and angles are equal.

Key Concepts

- **Criteria for Triangle Congruence:**

- SSS (Side-Side-Side): If three sides of one triangle are equal to three sides of another triangle, the triangles are congruent.
- SAS (Side-Angle-Side): If two sides and the included angle of one triangle are equal to two sides and the included angle of another triangle, the triangles are congruent.
- ASA (Angle-Side-Angle): If two angles and the included side of one triangle are equal to two angles and the included side of another triangle, the triangles are congruent.
- AAS (Angle-Angle-Side): If two angles and a non-included side of one triangle are equal to two angles and a non-included side of another triangle, the triangles are congruent.
- HL (Hypotenuse-Leg): For right triangles, if the hypotenuse and one leg are equal, the triangles are congruent.

Example Explanation

Consider two triangles with sides $a = 3, b = 4, c = 5$ and $a' = 3, b' = 4, c' = 5$. Using SSS, these triangles are congruent.

Practice Problems

1. Prove that triangles with sides 5, 12, 13 and 5, 12, 13 are congruent.
2. Show that two triangles with $\angle A = 50^\circ$, $\angle B = 60^\circ$, and $AB = 8$ are congruent.
3. Verify congruence for right triangles with hypotenuse 10 and one leg 6.
4. Are triangles with $\angle A = 70^\circ$, $AB = 5$, and $AC = 7$ congruent to triangles with $\angle A = 70^\circ$, $AB = 5$, and $AC = 7$?
5. Prove congruence of triangles with $\angle A = 90^\circ$, $AB = 7$, and $BC = 24$.

Solutions

1. By SSS, both triangles are congruent.
2. Using ASA, the triangles are congruent.
3. By HL, the triangles are congruent.
4. Using AAS, the triangles are congruent.
5. By HL, the triangles are congruent.

2 Similar Triangles

Triangles are similar if their corresponding angles are equal and their corresponding sides are proportional.

Key Concepts

- **Criteria for Triangle Similarity:**

- AA (Angle-Angle): If two angles of one triangle are equal to two angles of another triangle, the triangles are similar.
- SSS (Side-Side-Side): If the sides of one triangle are proportional to the sides of another triangle, the triangles are similar.
- SAS (Side-Angle-Side): If two sides are proportional and the included angle is equal, the triangles are similar.

Example Explanation

Consider triangles with sides $a = 3, b = 4, c = 5$ and $a' = 6, b' = 8, c' = 10$. The sides are proportional ($3 : 6, 4 : 8, 5 : 10$), so the triangles are similar by SSS.

Practice Problems

1. Prove that triangles with sides 2, 3, 4 and 4, 6, 8 are similar.
2. Show that two triangles with $\angle A = 30^\circ$, $\angle B = 60^\circ$, and $AB : BC = 1 : 2$ are similar.
3. Verify similarity for triangles with $\angle A = 45^\circ$, $\angle B = 45^\circ$, and $\angle C = 90^\circ$.
4. Are triangles with $\angle A = 80^\circ$ and sides $AB = 5$, $BC = 10$ similar to triangles with $\angle A = 80^\circ$, $AB = 10$, and $BC = 20$?
5. Prove similarity for triangles with $\angle A = 90^\circ$, $AB : BC = 3 : 4$, and $AB = 6$.

Solutions

1. By SSS, both triangles are similar.
2. Using AA, the triangles are similar.
3. By AA, the triangles are similar.
4. Using SSS, the triangles are similar.
5. By SAS, the triangles are similar.

3 Medians and Angle Bisectors

Medians and angle bisectors are important concepts in geometry with unique properties and applications.

Medians of a Triangle

A median of a triangle is a line segment joining a vertex to the midpoint of the opposite side. The medians of a triangle intersect at a single point called the centroid.

Key Properties: - The centroid divides each median into two parts, with the ratio 2 : 1 (the part closer to the vertex is twice as long as the part closer to the midpoint). - The centroid is the center of gravity of the triangle.

Angle Bisectors of a Triangle

An angle bisector divides an angle into two equal parts. The angle bisectors of a triangle meet at a single point called the incenter, which is the center of the inscribed circle (incircle).

Key Properties: - The incenter is equidistant from all three sides of the triangle. - The ratio in which the angle bisector divides the opposite side is proportional to the adjacent sides.

Practice Problems

1. Find the coordinates of the centroid of a triangle with vertices $(0, 0), (6, 0), (0, 8)$.
2. A triangle has sides $AB = 10$ and $AC = 6$. The angle bisector of $\angle BAC$ meets BC at D . Find the ratio $BD : DC$.
3. Prove that the three medians of a triangle always intersect at a single point.
4. Show that the incenter of a triangle is equidistant from all three sides.
5. In a triangle, the length of a median is 7. If the distance from the centroid to the opposite vertex is $\frac{14}{3}$, verify the ratio property of medians.

Solutions

1. Centroid is at $(2, \frac{8}{3})$.
2. Ratio $BD : DC = 5 : 3$ (angle bisector property).
3. Intersection proven using coordinate geometry or vector addition.

4. Use the property of angle bisectors and distance formula.
5. The ratio 2 : 1 is verified.

4 Theorems in Circles

Circles have unique properties governed by important theorems.

Key Theorems:

1. **Inscribed Angle Theorem**: The angle subtended by a chord at the center is twice the angle subtended on the circumference.
2. **Tangent-Secant Theorem**: If a tangent and a secant are drawn from an external point, $(\text{tangent segment})^2 = (\text{external part of secant}) \times (\text{whole secant})$.
3. **Power of a Point Theorem**: The power of a point is equal for tangents, chords, or secants passing through the point.
4. **Cyclic Quadrilateral Theorem**: Opposite angles of a cyclic quadrilateral sum to 180° .
5. **Equal Chords Theorem**: Chords equidistant from the center are equal in length.

Practice Problems

1. Prove the inscribed angle theorem using a triangle and isosceles properties.
2. In a circle, a tangent is drawn from a point P such that $PA = 5$ and $PBC = 15$. Verify the tangent-secant theorem.
3. Show that the opposite angles of a cyclic quadrilateral are supplementary.
4. A circle has chords $AB = 8$ and $CD = 8$. Prove they are equidistant from the center.
5. In a cyclic quadrilateral, $\angle A = 80^\circ$ and $\angle C = x$. Find x .

Solutions

1. Proven using isosceles triangles and supplementary angles.
2. Verified using the tangent-secant theorem formula.
3. Shown using cyclic properties and supplementary rules.
4. Proved using perpendicular distances from the center.
5. $\angle C = 100^\circ$ (cyclic quadrilateral theorem).